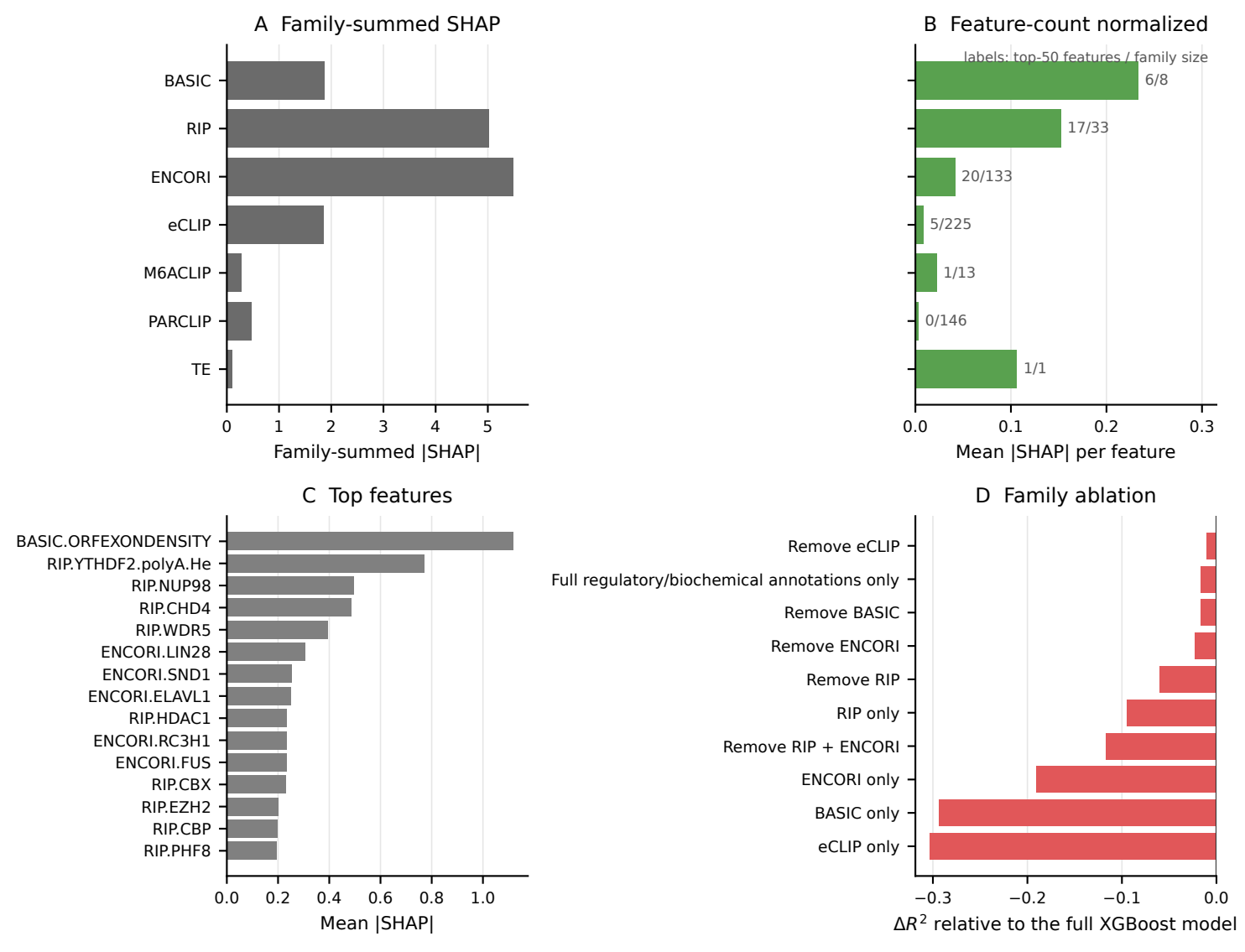
A Family-summed SHAP
BASIC
RIP
ENCORI
eCLIP
M6ACLIP
PARCLIP
TE
0
1
2
3
4
5
Family-summed |SHAP|
B Feature-count normalized
labels: top-50 features / family size
6/8
17/33
20/133
5/225
1/13
0/146
1/1
0.0
0.1
0.2
0.3
Mean |SHAP| per feature
C Top features
BASIC.ORFEXONDENSITY
RIP.YTHDF2.polyA.He
RIP.NUP98
RIP.CHD4
RIP.WDR5
ENCORI.LIN28
ENCORI.SND1
ENCORI.ELAVL1
RIP.HDAC1
ENCORI.RC3H1
ENCORI.FUS
RIP.CBX
RIP.EZH2
RIP.CBP
RIP.PHF8
0.0
0.2
0.4
0.6
0.8
1.0
Mean |SHAP|
D Family ablation
Remove eCLIP
Full regulatory/biochemical annotations only
Remove BASIC
Remove ENCORI
Remove RIP
RIP only
Remove RIP + ENCORI
ENCORI only
BASIC only
eCLIP only
−0.3
−0.2
−0.1
0.0
ΔR² relative to the full XGBoost model